

K2 Physics Validation Report

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Excerpt: passing benchmarks only. 1 failing benchmark omitted (ONERA M6 wing surface Cp vs experiment); see the full report for it.

18 passed · 0 failed · 1 skipped (49 gated comparisons)

4 further rows are marked **diag**: they record a known gap between a method and its reference — a low-order method used outside its own envelope — rather than asserting agreement. Diagnostic rows carry deliberately wide bands, cannot fail a benchmark, and are **not counted as passes** above.

Each engine is benchmarked against an independent reference, in three tiers of increasing strength:

1. **Exact solutions** — the 6DOF integrator against closed-form ODE solutions, structures against textbook formulas, the cone solver against Taylor-Maccoll.
2. **Code-to-code** — flight against OpenRocket, aerodynamics against SU2, the airframe FEM against CalculiX.
3. **Published measurement** — 2 benchmarks compare K2 against wind-tunnel data from the open literature. Agreement with another program proves consistency; agreement with a measurement is the only tier that can show the physics is right. The table below lists these alongside the other published references (exact-solution tables, agreed benchmark target values), which are external but are not measurements.

Comparisons against published data

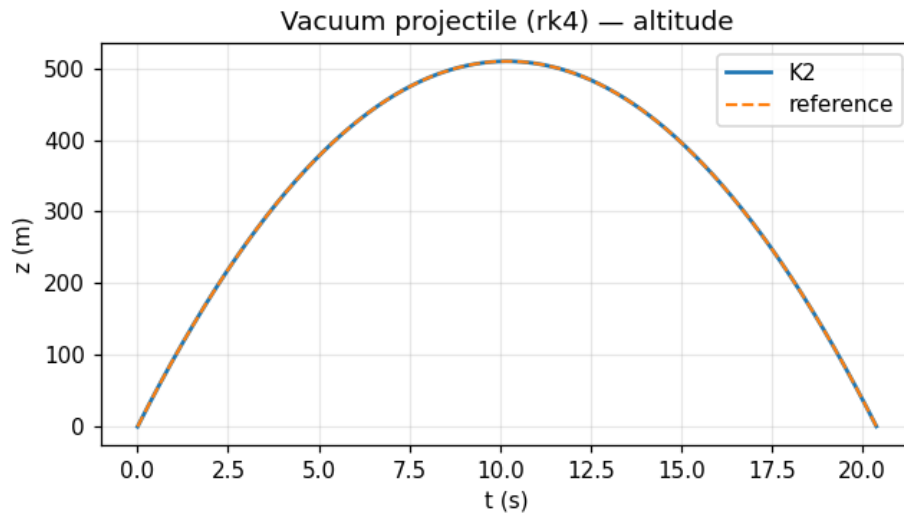
Benchmark	Source	Kind	Status
Taylor-Maccoll cone vs NACA-1135	NACA Report 1135 cone tables	published exact-solution tables	PASS
Barrowman lift slope vs AGARD-B experiment	AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T)	wind-tunnel measurement	PASS
SU2 lift slope vs AGARD-B experiment	AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T)	wind-tunnel measurement	PASS
NAFEMS LE1 — elliptic membrane (CalculiX)	NAFEMS standard benchmark target value	agreed benchmark target value	PASS
NAFEMS LE10 — thick plate under pressure (CalculiX)	NAFEMS standard benchmark target value	agreed benchmark target value	PASS

Flight Simulation (6DOF) ↔ Integrator exact solutions / OpenRocket

PASS — Vacuum projectile (rk4)

Reference: Exact kinematics $z=v_0\cdot t-\frac{1}{2}g\cdot t^2$ | Credibility: Validated

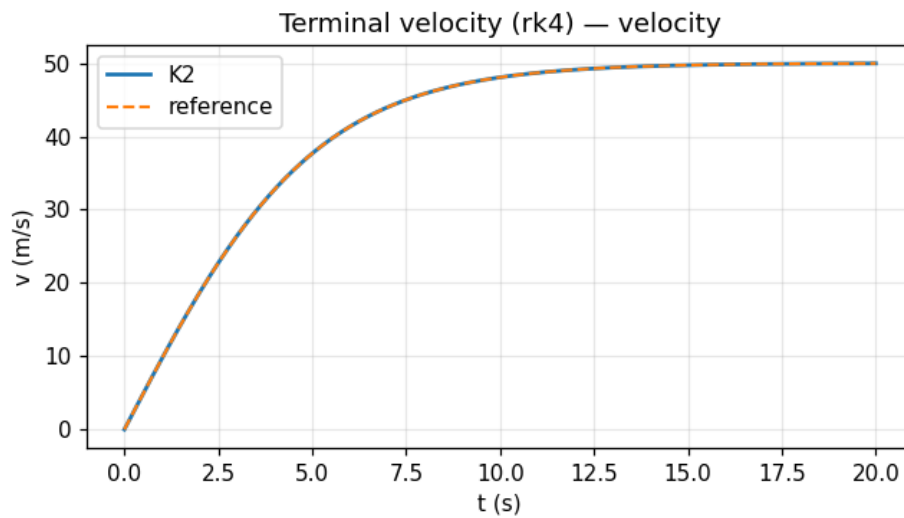
Quantity	K2	Reference	Source	Rel. err	Tol	Status
Apogee height	509.9	509.9	$v_0^2/2g$	0.00%	0%	✓
Position RMSE vs parabola	1.492e-11	0	exact	0.00%	/ 0.05	✓



PASS — Terminal velocity (rk4)

Reference: Exact $v(t)=v_t \cdot \tanh(g \cdot t / v_t)$, $v_t = \sqrt{g/k}$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Terminal velocity	49.96	50	$\sqrt{g/k}$	0.08%	0%	✓
Velocity RMSE vs tanh	5.968e-12	0	exact	0.00%	/ 0.02	✓



PASS — Oscillator energy (rk4)

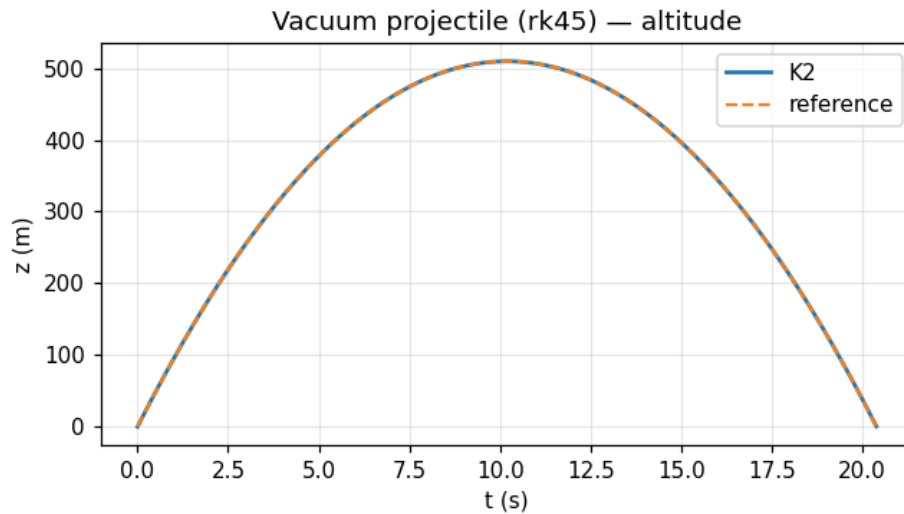
Reference: Conserved $E = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 x^2$ over 50 periods | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Energy drift (50 periods)	2	2	$E(0)$	0.00%	1%	✓

PASS — Vacuum projectile (rk45)

Reference: Exact kinematics $z = v_0 \cdot t - \frac{1}{2}g \cdot t^2$ | Credibility: Validated

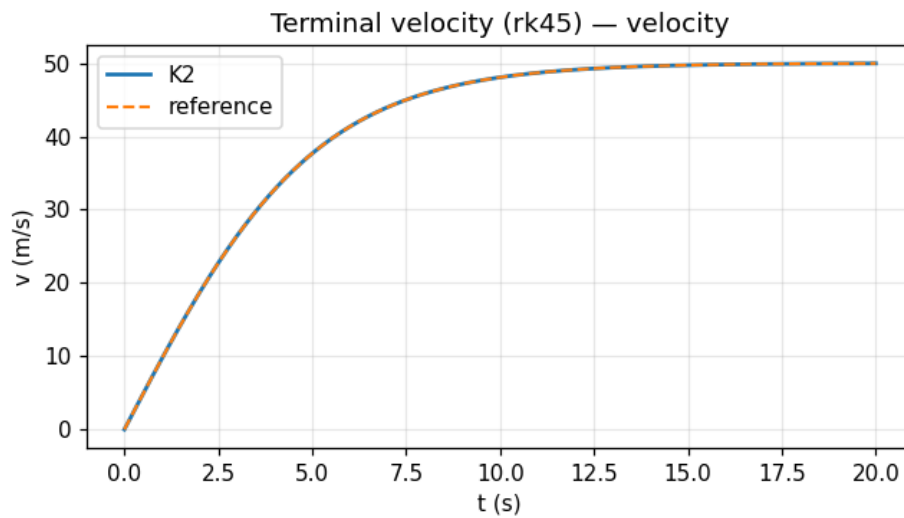
Quantity	K2	Reference	Source	Rel. err	Tol	Status
Apogee height	509.9	509.9	$v_0^2/2g$	0.00%	0%	✓
Position RMSE vs parabola	1.493e-11	0	exact	0.00%	/ 0.05	✓



PASS — Terminal velocity (rk45)

Reference: Exact $v(t)=v_t \cdot \tanh(g \cdot t / v_t)$, $v_t = \sqrt{g/k}$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Terminal velocity	49.96	50	$\sqrt{g/k}$	0.08%	0%	✓
Velocity RMSE vs tanh	1.423e-13	0	exact	0.00%	/ 0.02	✓



PASS — Oscillator energy (rk45)

Reference: Conserved $E = \frac{1}{2}v^2 + \frac{1}{2}\omega^2 x^2$ over 50 periods | Credibility: Validated

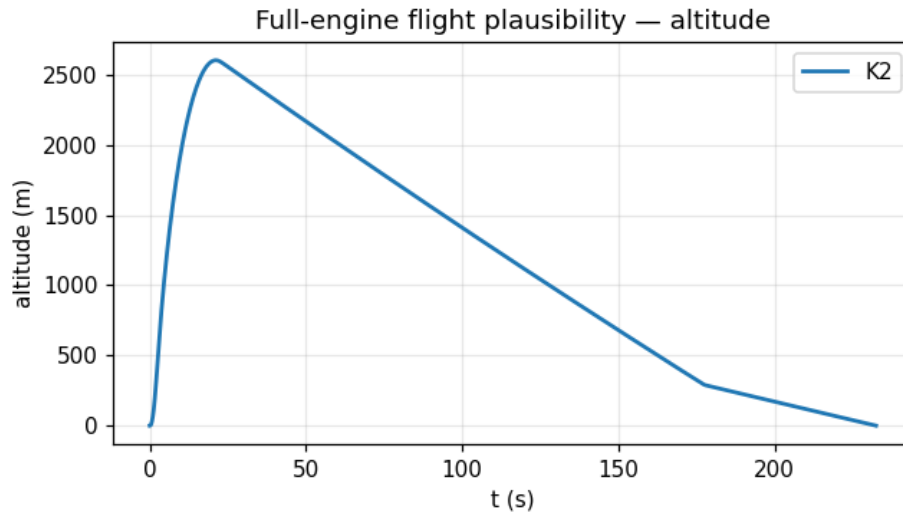
Quantity	K2	Reference	Source	Rel. err	Tol	Status
Energy drift (50 periods)	2	2	$E(0)$	0.00%	1%	✓

PASS — Full-engine flight plausibility

Reference: Physical sanity bounds (canonical L1090W rocket) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Apogee in [500, 6000] m	1	1	bound	0.00%	/ 0.5	✓
Vehicle landed	1	1	bound	0.00%	/ 0.5	✓

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Max-Q before apogee	1	1	bound	0.00%	/ 0.5	✓
Subsonic-to-low-supersonic max Mach	1	1	bound	0.00%	/ 0.5	✓



Sim vs OpenRocket (skipped)

Reference: OpenRocket flight CSV (canonical rocket)

Reason: set OPENROCKET_CSV to an OpenRocket flight-data CSV of the canonical rocket (build it per validation/cases/rocket_canonical.py, motor L1090W, and export flight data)

Aerodynamics ↔ Wind-tunnel measurement / Taylor-Maccoll exact / SU2

PASS — Taylor-Maccoll cone vs NACA-1135

Reference: NACA Report 1135 cone tables | Credibility: Validated

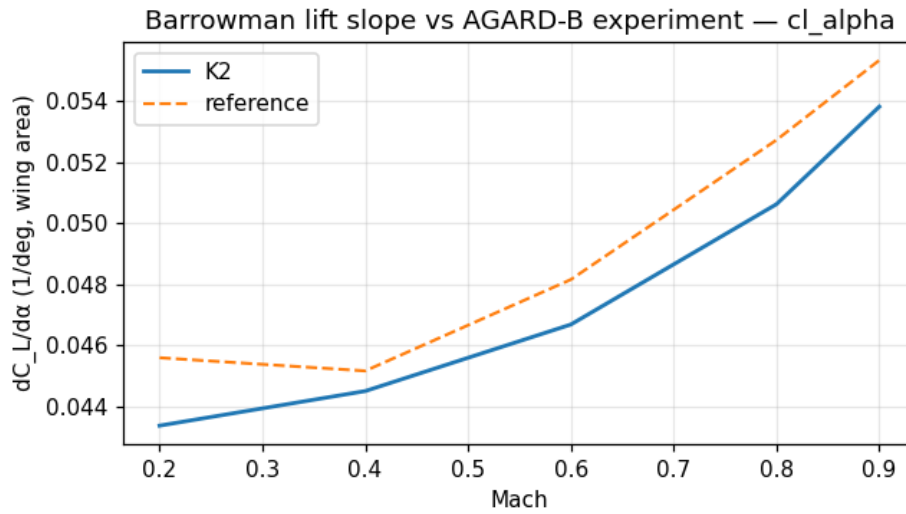
Quantity	K2	Reference	Source	Rel. err	Tol	Status
Shock angle ($M=2.0$, $\theta_c=10.0^\circ$)	31.21	31.2	NACA-1135	0.02%	1%	✓
Surface C_p ($M=2.0$, $\theta_c=10.0^\circ$)	0.1045	0.105	NACA-1135	0.50%	5%	✓
Shock angle ($M=3.0$, $\theta_c=10.0^\circ$)	21.71	21.8	NACA-1135	0.39%	1%	✓
Surface C_p ($M=3.0$, $\theta_c=10.0^\circ$)	0.08748	0.088	NACA-1135	0.59%	5%	✓

PASS — Barrowman lift slope vs AGARD-B experiment

Reference: AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
$C_{L\alpha}(M=0.4) / C_{L\alpha}(M=0.2)$	1.026	0.9906	AEDC-TR-70-100	3.59%	8%	✓
$C_{L\alpha}(M=0.6) / C_{L\alpha}(M=0.2)$	1.076	1.056	AEDC-TR-70-100	1.91%	8%	✓
$C_{L\alpha}(M=0.8) / C_{L\alpha}(M=0.2)$	1.167	1.156	AEDC-TR-70-100	0.93%	8%	✓
$C_{L\alpha}(M=0.9) / C_{L\alpha}(M=0.2)$	1.241	1.214	AEDC-TR-70-100	2.24%	8%	✓
$C_{L\alpha}$ at $M=0.2$ (absolute)	0.04338	0.0456	AEDC-TR-70-100	4.88%	10%	✓
$C_{L\alpha}$ at $M=0.6$ (absolute)	0.04669	0.04816	AEDC-TR-70-100	3.06%	10%	✓
$C_{L\alpha}$ at $M=0.9$ (absolute)	0.05382	0.05534	AEDC-TR-70-100	2.75%	10%	✓
C_{D0} at $M=0.2$ (diagnostic)	0.03097	0.02583	AEDC-TR-70-100	19.92%	10%	diag

Quantity	K2	Reference	Source	Rel. err	Tol	Status
C_D0 at M=0.5 (diagnostic)	0.03457	0.02609	AEDC-TR-70-100	32.49%	10%	diag
C_D0 at M=0.9 (diagnostic)	0.05163	0.02802	AEDC-TR-70-100	84.25%	10%	diag



PASS — SU2 cone vs Taylor-Maccoll

Reference: Taylor-Maccoll exact (M=2, 10° cone) | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Cone surface Cp (M=2, 10°)	0.09942	0.1045	Taylor-Maccoll	4.84%	10%	✓

PASS — Barrowman aero vs SU2

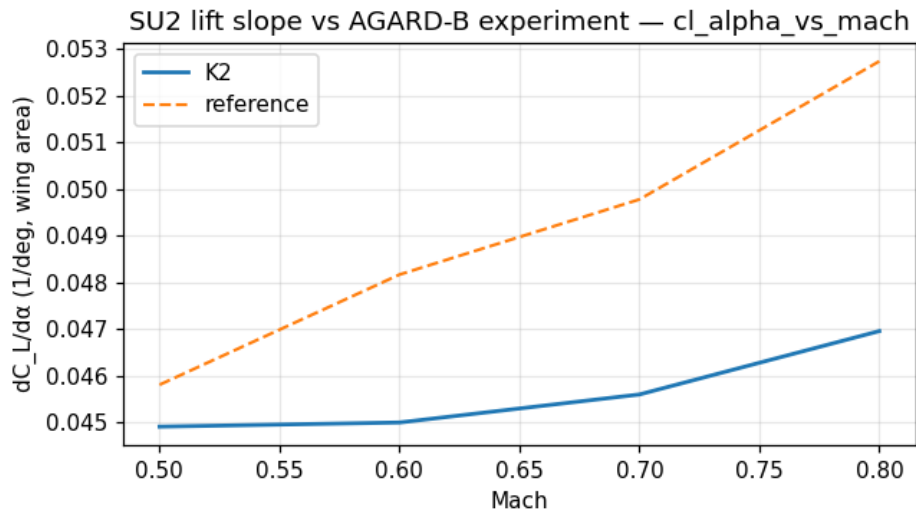
Reference: SU2 RANS (canonical rocket) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
CFD reference area	0.008171	0.008171	$\pi \cdot (d_{\text{body}}/2)^2$	0.00%	2%	✓
Drag positive	1	1	sign	0.00%	/ 0.5	✓
Lift positive at +AoA	1	1	sign	0.00%	/ 0.5	✓
Normal force Cn (body axes)	1.432	1.327	SU2	7.88%	10%	✓
Drag coefficient Cd (wind axes)	0.4548	0.4046	SU2	12.41%	10%	diag

PASS — SU2 lift slope vs AGARD-B experiment

Reference: AEDC-TR-70-100 wind-tunnel data (AGARD-B, Tunnel 4T) | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
$dC_L/d\alpha$ at M=0.5	0.0449	0.0458	AEDC-TR-70-100	1.96%	15%	✓
C_L at M=0.5, $\alpha=6^\circ$	0.2686	0.2761	AEDC-TR-70-100	2.74%	20%	✓
$dC_L/d\alpha$ at M=0.6	0.04499	0.04816	AEDC-TR-70-100	6.58%	15%	✓
C_L at M=0.6, $\alpha=6^\circ$	0.2693	0.2934	AEDC-TR-70-100	8.23%	20%	✓
$dC_L/d\alpha$ at M=0.7	0.04559	0.04977	AEDC-TR-70-100	8.40%	15%	✓
C_L at M=0.7, $\alpha=6^\circ$	0.2731	0.3038	AEDC-TR-70-100	10.10%	20%	✓
$dC_L/d\alpha$ at M=0.8	0.04695	0.05273	AEDC-TR-70-100	10.96%	15%	✓
C_L at M=0.8, $\alpha=6^\circ$	0.2817	0.3217	AEDC-TR-70-100	12.42%	20%	✓



Structures ↔ NAFEMS benchmarks / textbook closed form / CalculiX

PASS — Closed-form structural formulas

Reference: Textbook thin-wall / Euler relations | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Hoop stress σ_h	4e+07	4e+07	$p \cdot r/t$	0.00%	2%	✓
Euler buckling P_{cr}	2.752e+05	2.752e+05	$\pi^2 EI/L^2$	0.00%	5%	✓
von Mises (uniaxial)	1.5e+08	1.5e+08	σ	0.00%	0%	✓

PASS — Uniaxial bar (CalculiX)

Reference: Exact $\delta = FL/EA$, $\sigma = F/A$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Tip elongation δ	1.797e-05	1.814e-05	FL/EA	0.95%	3%	✓
Axial stress σ (mean)	2.5e+06	2.5e+06	F/A	0.00%	2%	✓

PASS — Cantilever bending (CalculiX)

Reference: Euler-Bernoulli $\delta = FL^3/3EI$ | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
Tip deflection δ	0.002242	0.002268	$FL^3/3EI$	1.11%	5%	✓

PASS — 1st bending mode: closed-form vs CalculiX

Reference: CalculiX modal (cantilever) | Credibility: Estimated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
f_1 (1st lateral bending)	25.81	25.78	CalculiX mode 1	0.11%	25%	✓

PASS — NAFEMS LE1 — elliptic membrane (CalculiX)

Reference: NAFEMS standard benchmark target value | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
σ_{yy} at point D	9.294e+07	9.27e+07	NAFEMS	0.26%	5%	✓
σ_{yy} at D (integration points, diagnostic)	7.949e+07	9.27e+07	NAFEMS	14.25%	40%	✓

PASS — NAFEMS LE10 — thick plate under pressure (CalculiX)

Reference: NAFEMS standard benchmark target value | Credibility: Validated

Quantity	K2	Reference	Source	Rel. err	Tol	Status
σ_{yy} at point D	-5.596e+06	-5.38e+06	NAFEMS	4.01%	10%	✓
σ_{yy} at D (integration points, diagnostic)	-4.282e+06	-5.38e+06	NAFEMS	20.40%	40%	✓